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Mini Project Proposal - BAI506

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AI Loan Underwriting Assistant: A Multi-Agent Agentic AI System for Faster Loan Processing			
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Signature of Guide

Initial Project Proposal

Loan processing in banks involves reviewing applicant documents, extracting financial information, checking eligibility, assessing repayment risk, identifying inconsistencies, and preparing an underwriting recommendation. These activities can require substantial manual effort because relevant information is distributed across multiple documents. This project proposes an Agentic Artificial Intelligence (AI) based Loan Underwriting Assistant that assists bank loan officers by coordinating multiple specialized agents. A Document Intake Agent (DIA) extracts structured information from applicant documents using Optical Character Recognition (OCR) and document understanding. Eligibility and Risk Agents (RA) evaluate predefined lending conditions and repayment-related indicators, while a Fraud/Anomaly Agent checks for inconsistencies across documents. A Decision and Reasoning Agent (DRA) combines the findings and generates an explainable recommendation of approve, reject, or manual review. The recommendation is presented through a dashboard where the loan officer reviews the evidence and makes the final decision. The prototype will use synthetic applicant profiles and sample documents to avoid exposing real customer information. The proposed workflow aims to reduce repetitive manual work, speed up preparation of underwriting information, apply rules consistently, and provide a clear reasoning summary and audit record for each processed application.

1. Introduction

Loan underwriting is the process of evaluating a loan application using information about the applicant's identity, income, existing obligations, financial records, and repayment capacity. In a conventional workflow, a loan officer may need to examine several documents, extract relevant values, compare information across documents, perform financial checks, identify exceptions, and prepare a recommendation. Our project proposes a multi-agent Agentic AI system that acts as a Loan Underwriting Assistant. The system divides the underwriting-support workflow into specialized tasks and coordinates them through an Orchestrator Agent (OA). The purpose is to reduce repetitive processing, not to replace the authorized loan officer and present the relevant evidence and recommendation in a consolidated form.

1.1 Motivation

The motivation for this project is to help banks process loan applications faster while retaining human oversight. Research has demonstrated that machine learning can improve the processing of lending information and that explainable models can provide reasons for credit-risk decisions. Research in India and Singapore also highlights the potential of artificial intelligence and machine learning to improve lending efficiency, while emphasizing issues such as transparency, data quality, and responsible use. The proposed project combines these ideas with a multi-agent workflow so that document processing, eligibility analysis, risk assessment, anomaly detection, and recommendation generation can be coordinated as one loan-processing support system.

2. Literature Survey

This survey considers research from the United States, Europe, India, and Singapore/ASEAN covering explainable credit-risk assessment, human information processing in lending, automated loan-risk assessment, machine-learning based credit scoring, and multi-agent systems. The given table states the specific work being referenced and how it supports the proposed project.

Sl. No.	Paper / Author	What is being referenced	Relevance to proposed project
1	W. Wang et al., "Using Small Business Banking Data for Explainable Credit Risk Scoring," AAAI, 2020. [USA]	Credit-risk scoring and simple explanations using banking data.	Supports explainable risk assessment.
2	M. Liu, "Assessing Human Information Processing in Lending Decisions," Journal of Accounting Research, 2022. [USA]	Machine learning can process lending information while human officers add judgement.	Supports the human-in-the-loop Assistant.
3	N. Bussmann et al., "Explainable Machine Learning in Credit Risk Management," Computational Economics, 2021. [Europe]	Explainable machine learning for credit-risk decisions.	Supports clear reasons for risk results.
4	M. Tetteh et al., "Credit Risk Assessment with Madaline and Multilayer Perceptrons," Springer, 2024. [India]	Neural networks for automated loan-risk assessment.	Supports the Risk Agent.
5	P. Jutasompakorn, A. Perdana, and V. Balachandran, "Enhancing Decision Making with Machine Learning: The Case of Aurora Crowdlending Platform," 2023. [Singapore]	Machine learning for credit scoring and lending decisions.	Supports data-driven loan assessment.
6	N. Remolina, "The Role of Financial Regulators in the Governance of Algorithmic Credit Scoring," SMU, 2022. [Singapore]	Governance, privacy, fairness, and oversight in algorithmic credit scoring.	Supports human review and auditability.
7	Q. Wu et al., "AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation," ICLR Workshop, 2024.	Multiple language-model agents can communicate and work together.	Supports the multi-agent architecture.

The literature shows that machine learning can support credit-risk assessment, explainable methods can make results easier to understand, and human judgement remains important in lending. Multi-agent research also shows that specialized agents can work together on complex tasks. Our project combines these ideas into one loan-processing Assistant.

2.1 Limitations of Existing System

- Real banking deployment requires careful handling of privacy, data quality, fairness, explainability, and regulatory requirements.
- Manual processing requires loan officers to inspect and consolidate information from several applicant documents.
- Conventional credit-risk models may provide a score without presenting all the evidence in a form that is easy for a loan officer to review.
- Document extraction, eligibility checking, risk assessment, and anomaly checking are usually separate activities and might require repeated manual coordination.
- Exceptional or inconsistent applications still need human judgement and cannot be safely handled only through an automated score.

2.2 Outcome of the Literature Survey

The main gap: document processing, risk analysis, anomaly checking, and explanation are usually handled separately. Our project combines these tasks through multiple agents and presents one clear recommendation to the loan officer.

3. Problem Statement

To design and develop a multi-agent AI based Loan Underwriting Assistant that assists banks in processing loan applications faster by extracting information from applicant documents, evaluating eligibility and repayment risk, detecting inconsistencies, and generating an explainable loan recommendation for review and final decision by a human loan officer.

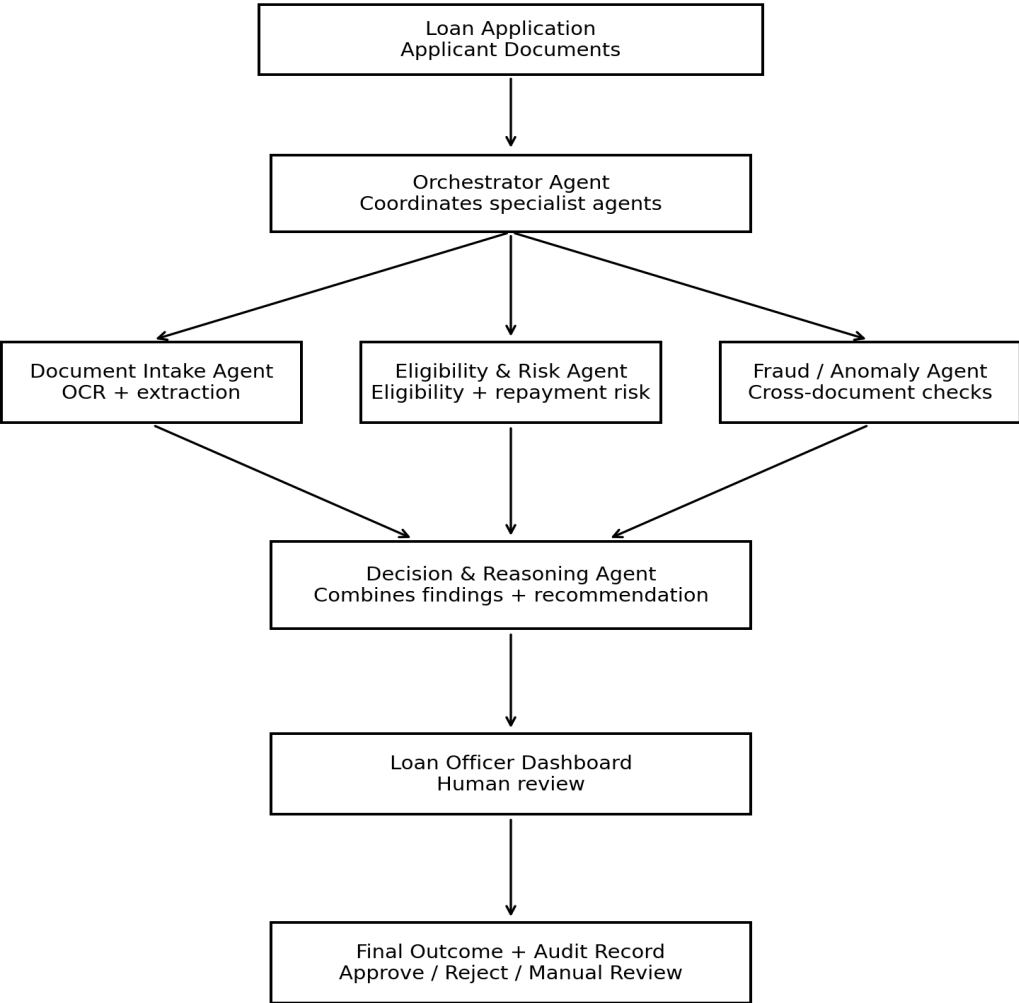
3.1 Objectives

- To develop a multi-agent AI workflow that automates document understanding, eligibility and risk analysis, and fraud/anomaly checking for loan applications.
- To generate an explainable loan-processing report and recommendation through a ‘human-in-the-loop dashboard’, thereby reducing repetitive manual effort and supporting faster underwriting.

4. Proposed Methodology

The proposed methodology begins when an applicant's loan documents are uploaded into the system. The OA manages the workflow and routes the required information to specialized agents. The DIA extracts relevant fields such as applicant details, income, liabilities, and other financial information using Optical Character Recognition and document-understanding techniques. The Eligibility Agent checks predefined lending conditions. The RA evaluates repayment-related indicators using the extracted information and predefined rules or models. The Fraud/Anomaly Agent compares information across documents and flags inconsistencies such as mismatched names, income differences, or unusual financial patterns.

The DRA receives the outputs of the specialist agents, combines the findings, and prepares a recommendation of Approve, Reject, or Manual Review together with the main reasons and supporting evidence. The recommendation is displayed on a loan officer dashboard. The loan officer reviews the extracted information, risk findings, anomaly flags, and reasoning before confirming or overriding the recommendation. The final outcome is recorded as an audit-oriented project record. Synthetic applicant data and sample documents will be used for the prototype so that real customer banking information is not required.



Proposed System Methodology

5. System Requirement Specifications (Technical & Non-Technical)

Non-Technical Requirement	Specification
Ease of use	The dashboard should be simple and clear.
Response time	The prototype should provide results with low delay.
Accuracy	Extracted data and risk results should be reliable.
Explainability	The system should show the main reasons for its recommendation.
Privacy	Use synthetic or anonymized applicant data.
Maintainability	Agents and rules should be easy to update.
Document input	Sample loan application documents such as identity proof, salary slips, bank statements and supporting financial documents in PDF or image format.
Programming environment	Python development environment for implementing the agents, data processing and application workflow.
Artificial Intelligence models	Large Language Model (LLM) for document understanding, agent reasoning and report generation; predefined rules/models for numerical eligibility and risk calculations.
Document processing	Optical Character Recognition (OCR) and document-understanding components for extracting structured applicant information.
Multi-agent orchestration	OA with specialized agents for document intake, eligibility, risk, fraud/anomaly checking, and decision reasoning.
Application interface	Streamlit or equivalent web interface for document upload, processing status, findings, recommendation and human review.
Data	Synthetic or anonymized applicant profiles and sample documents for development, testing and evaluation.
Hardware	Standard computer with sufficient processing capability and internet access when cloud-based Artificial Intelligence services are used.
Usability	The dashboard should present extracted information, findings and recommendations clearly so that a loan officer can review an application quickly.
Human oversight	The system should provide a recommendation only; the authorized loan officer should retain responsibility for the final decision.
Explainability	Each recommendation should be accompanied by the major evidence, eligibility/risk findings and anomaly flags that contributed to it.
Privacy	Real customer banking information should not be required for the prototype; synthetic or anonymized data should be used during development and evaluation.
Reliability	The system should identify missing, uncertain or inconsistent information rather than silently producing a complete recommendation.
Performance	The workflow should process a small application package within a practical time for demonstrating reduction in repetitive processing effort.
Maintainability	Agents, prompts, rules and processing modules should be modular so that individual components can be updated independently.
Responsible use	The prototype should be treated as decision-support software and should not be presented as an autonomous replacement for bank underwriting authority.

Technical Requirement	Specification
Document upload	Upload sample loan documents.
Data extraction	Extract applicant and financial details.
Agent coordination	Coordinate agents through the Orchestrator Agent.
Eligibility and risk	Check eligibility and repayment risk.
Fraud/anomaly check	Find mismatched or unusual information.
Recommendation	Generate Approve, Reject, or Manual Review recommendation.
Dashboard	Show findings and recommendation to the loan officer.

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